16 – Agents

Definition: An **agent** is an AI that is given a goal, makes its own plan, works through it step by step using tools, and checks intermediate results – instead of just writing a single answer.

> **Remember:** The chatbot answers. The agent acts.

Key points

- **The cycle:** goal → plan → action → observation → next action. The agent runs this loop until it considers the goal reached.
- **Tools:** The agent uses what topic 11 covers (web search, files, code) – and does so even when you're not watching.
- **Examples:** research agent (collects sources), coding agent (writes and tests code), scheduling or email assistant, browser agent (operates websites).
- **Strengths:** Automating multi-step tasks that would be tedious one at a time.
- **Weaknesses:** Errors accumulate (one wrong step distorts everything after it), costs from many steps, loss of control, security risks (topics 12, 21).
- **When not to:** for simple tasks (the chat is enough), for high-stakes steps without oversight (payments, publications).
- **Your role:** state the goal clearly, set boundaries, review intermediate results. The agent plans; you are accountable.

Example

Task: "For a short trip to Sample City, find three accommodations under €80 per night in a central location, and compare them in a table."

Step	What the agent does	What you check
1	Plan: search, filter, compare	Does the plan fit the goal?
2	Web search, open pages	Is it using reputable sources?
3	Collect results, check prices	Are the prices current?
4	Build the table	Do the numbers spot-check?

Something can go wrong at every step: outdated prices, a page that misleads the agent (topic 12). That's why: look at intermediate results, not just the final result.

Do / Don't

Do	Don't
State goals clearly and verifiably	Give vague orders ("Just take care of it")
Review intermediate steps and the result	Let agents run unsupervised on consequential tasks
Handle small tasks in the chat	Start an agent for every little thing

Safety note

An agent with tools can *do* things. Give it only the permissions the task requires, and require approvals before actions with consequences (topic 21).

Mini checklist

- ☐ I can explain the difference between a chatbot and an agent in one sentence.
- ☐ I know the cycle plan → action → observation.
- ☐ I know that agent errors accumulate.
- ☐ I review intermediate results, not just the final result.

Exercise (5 min.)

Take a task from your everyday life (e.g. planning a vacation, a weekly plan). Break it down into planning steps like in the example, and mark for each step: does a human need to check here? What goes wrong if not?

Next steps

- Extra exercises: `02_extra_exercises/16_extra_exercises`
- Test: Test E (Agentic systems)
- Related: 11 tools, 17 MCP, 19 subagents, 21 agent safety

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